

Electromagnetic shielding for the protection of crew in the space radiation environment

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Abstract

The Artemis program will send the next humans to the Moon as soon as 2024. In Low Earth Orbit, astronauts are protected by the Earth's magnetic field, however, in lunar orbit and on the lunar surface, the radiation environment is significantly different and brings associated health risks. Currently, all radiation protection is provided by passive shielding; the dimensions of which would need to be increased to provide the same protection in the higher radiation environment, however, increases in material thickness result in increased mass which becomes too heavy and bulky to be launched. Active shielding, from electromagnets, provide a lighter weight, alternative shield capable of deflecting charged particles away from spacecraft. Shields generating magnetic field strengths of < 10 mT have been constructed and tested in laboratory conditions. Simulations have shown that a shield with this design could deflect 1 keV protons by $\sim 8^\circ$ using a peak field strength of 72 mT.

Keywords: human spaceflight, ionising radiation, electromagnetic shielding, magnetism, Artemis.

Introduction

The 2024 Artemis missions will be the next time humans set foot on the moon. A large concern for the health and safety of these astronauts will be the exposure to high doses of ionising radiation, both on their way to the moon as well as whilst on the moon. Current forms of radiation shielding only provide passive methods of protection.

One of the major issues facing humans in long-term existence in space, is radiation. Radiation is a normal part of daily life on Earth, but we typically experience very small amounts of it over the course of one year. In space, the amount of radiation an astronaut on the International Space Station (ISS) is exposed to over a 6-month period can be upwards of 70 years equivalency back on Earth [1]. For a journey to Mars using the shortest calculated return trip time, and current technology, the dose equivalent expected to be absorbed by astronauts is just over 13 times NASA's annual limit of 50 mSv/yr [2]. According to the NASA Human Research Program, the long-term health risks of missions with duration > 1 yr, pose an unacceptable risk of cancer to astronauts and requires mitigation [3].

Space radiation is made of two types, solar radiation from the Sun and radiation from outside of our solar system, known as Galactic Cosmic Rays (GCR). The Sun is a blackbody radiator with majority of the energy emitted across the infrared and visible light range. Solar events, such as solar flares or Coronal Mass ejections (CME) cause large amounts of charged particles, in the form of electrons, ions and protons to be ejected from the Sun with high energies. GCR radiation is approximately 98% nuclei, which is made of about 87% Hydrogen nuclei, 12% Helium nuclei and 1% nuclei of heavier elements, known as HZE ions [4].

Currently the main form of protection from radiation is by means of passively absorbing the energy in large quantities of dense materials. The most popular material for passive space

radiation shielding is Polyethylene (PE), a common polymer that makes up many plastic goods. PE is a good absorber of space radiation due to its high Hydrogen content [5].

The issue with passive shielding techniques is that they only protect the area directly behind the incident radiation. For an entire module solution, shielding must be manufactured, and then launched into space, such that it can provide shielding to large portions of the areas occupied by astronauts. As the cost of shipping materials into space is very expensive, launching large quantities of these shields is not feasible for long-term missions to Mars, or for extra-terrestrial bases.

Active radiation may serve as a lighter, cheaper alternative to current passive methods of radiation protection. Research has previously been performed into active radiation shielding using electrostatic and magnetic shields to completely block charged space radiation [6] but none have been implemented on spacecraft. The Alpha Magnetic Spectrometer currently on the ISS is a 6.7 tonne electromagnet used to measure radiation, but it provides no shielding of crew inside.

Methods

Magnetic Field Model

Creating a magnetic field requires current to flow within a conductor. The strength of the field generated depends on the amount of current flowing. This relationship is determined by the Biot-Savart law, for DC currents; where B is magnetic field strength, I is current, L is the length of the current-carrying wire and r is the distance from the wire.

$$\vec{dB} = \frac{\mu I d\vec{L} \hat{r}}{4\pi r^2} \quad (1)$$

The strength of the magnetic field drops off inversely proportional to the square of the distance. This results in the need for either a large current, or many wire lengths close together to produce a strong field. Using a coil will provide a strong central magnetic field down the length, due to the many wire lengths in close proximity. Coils have the added benefit of keeping their strong magnetic field contained within their centre, with minimal field outside. Reducing the magnetic field spillage into the protected area is optimal, as strong magnetic fields have negative effects on humans and electronics.

The deflection capability of a magnetic field depends on the strength and size of the field. In the case of the Earth, the magnetic field is relatively small, ranging between 25 to 65 μT at the surface, but is massive in scale, reaching thousands of kilometres into space. The sheer size of the Earth's protective magnetic field allows it to have a prolonged effect on any charged particles on a trajectory towards Earth. In the case of a spacecraft, the magnetic field would be limited to a much smaller size, but can produce a stronger field, within the limits of the power system on board.

Only charged particles experience a force when they travel through a magnetic field, provided they are not travelling parallel with the field. This interaction can be defined through the Lorentz force equation, which also takes into consideration the coulombic force between two charges.

$$\vec{F} = q\vec{E} + q\vec{v}\vec{B} \quad (2)$$

A MATLAB script was created to model the interaction between a charged particle and a magnetic field in a 3D space environment. This script implemented the Lorentz force equation for magnetic interactions, assuming no interactions between charged particles took place.

Computer-Aided Design (CAD)

The Destiny ISS module was used as a reference for the scaled down CAD design. The side walls were replaced with several PE sheets to act as the passive shielding element. A copper coil was then modelled around each, to simulate the active shielding component. A structural shell was also designed to simulate structural supports for the module, as well as mounting of the panels and other necessary wall materials.

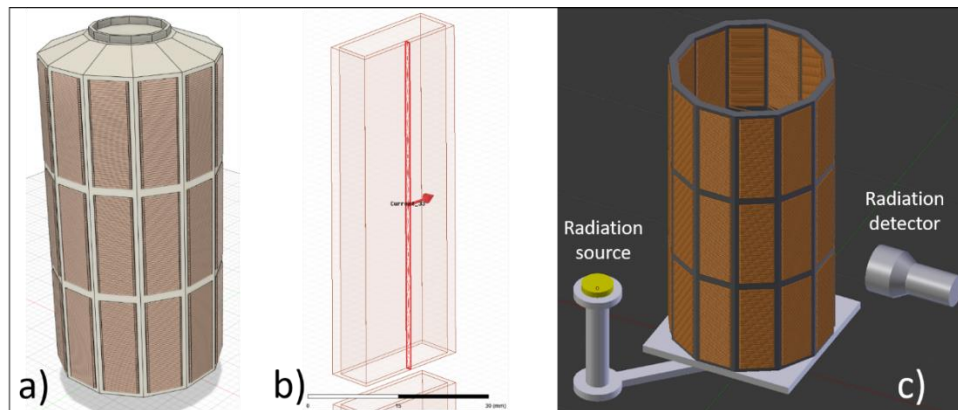


Fig. 1: CAD models of spacecraft modules with electromagnetic shields in the walls of the spacecraft. (a) full module (b) individual electromagnet (direction of magnetic field indicated) and (c) radiation testing experimental configuration.

Shield Prototypes

Two shields have thus far been constructed with the field-in-wall design. Figure 2 shows photographs of the two shields constructed with (a) 10 cm diameter and (b) 6 cm diameter. The smaller shield has been constructed specifically to fit inside a 1U payload for potential launch on a high-altitude sounding rocket where space and weight are limited.

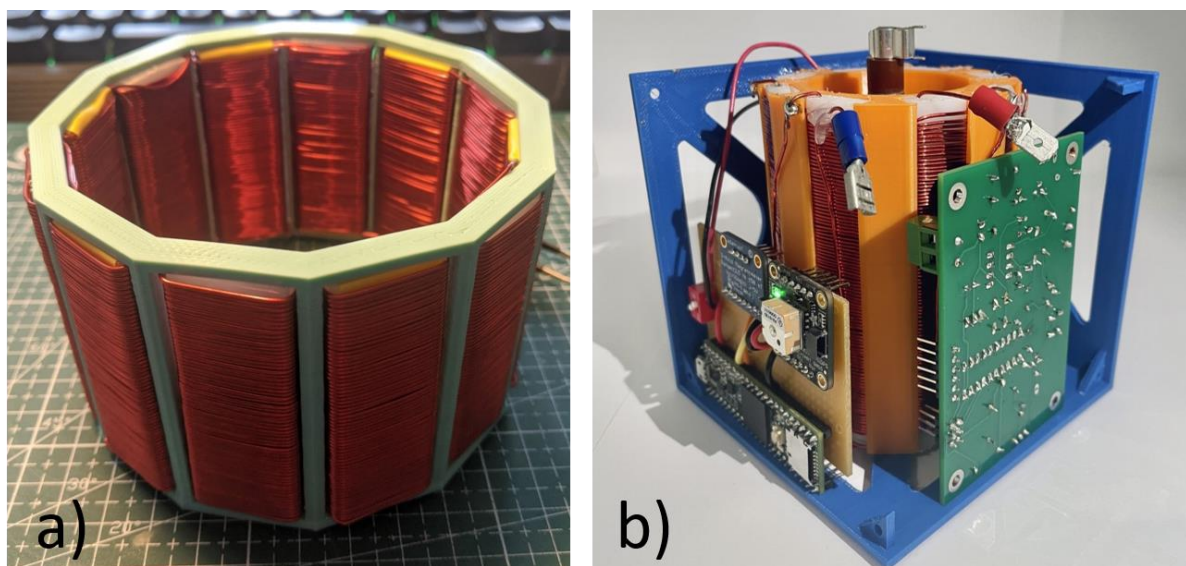


Fig. 2: Electromagnetic shield prototypes constructed by placing the magnetic fields in the wall of the shield. (a) simple shield, 10 cm diameter, 12 sections, $B = 40$ mT. (b) 1U shield, 6 cm diameter, 8 sections, $B = 10$ mT.

Results

Magnetic Fields (Simulations)

Simulations using MATLAB have generated particle deflections of varying angles depending on energy of the incoming particles. Figure 3 provides the plots of the deflections for a circular shield with the magnetic field in the walls of the shield.

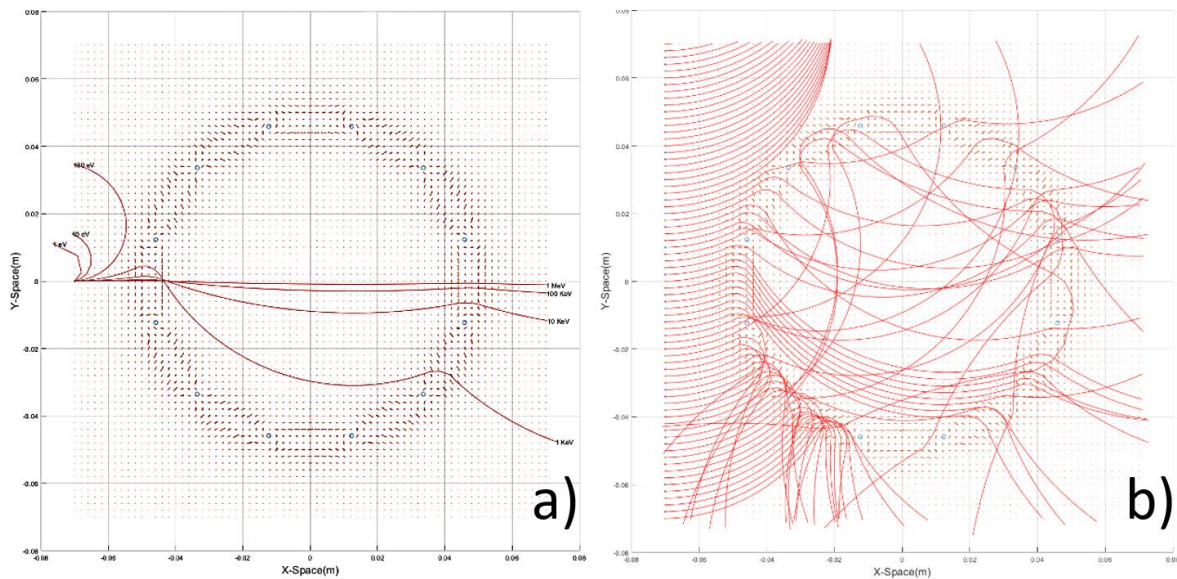


Fig. 3: 1 T peak magnetic field deflecting protons with (a) a range of energies; (b) 1 keV protons.

Magnetic Fields (Experiments)

Power is provided using a mains power supply with control over both voltage and current. Magnetic field strength was measured using the probe of a Gauss meter positioned at the end of each polyethylene slab. Table 1 provides the resulting magnetic fields with varying input current, as per the relationship given in equation 1.

Table 1: Experimental data obtained from benchtop measurements on the prototype shield.

PSU voltage (V)	Measured voltage (V)	I (A)	B (mT)
2	1.46	1.12	0.84
4	2.99	2.30	1.04
6	4.37	3.36	1.27
8	5.87	4.52	1.54
10	7.46	5.74	1.80
12	8.78	6.75	2.00

Particle Deflection

A field strength of 1 T (100 times stronger than the Earth's magnetic field) is capable of deflecting 1 keV protons by 30°. For a 4 m diameter habitable spacecraft module, such a deflection is sufficient to bypass the central location of the module. Figure 4 illustrates the particle trajectories after interaction with the outer hull of the spacecraft.

As particle energies are increased, so too must the strength of the magnetic field. Herein lies the challenge: the ISS can deliver a maximum of 160 V voltage from the solar panels, but this is first regulated to reliably deliver 120 V or 60 V to equipment on board. To generate the high

currents required to generate magnetic fields greater than 1 T, some innovative electrical circuitry is required.

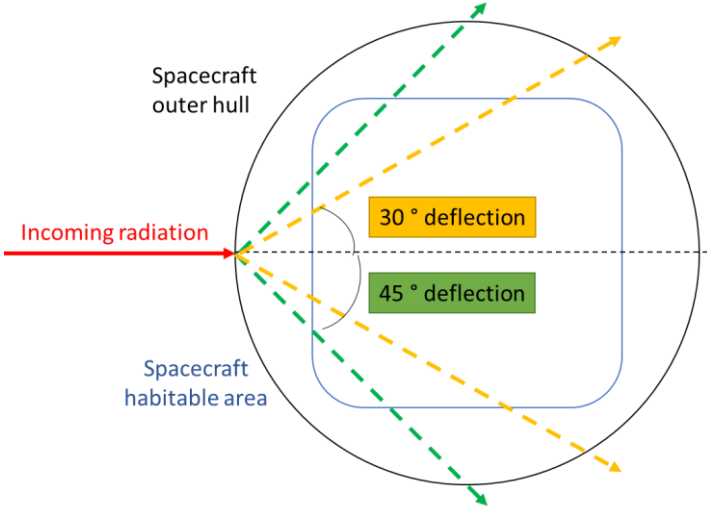


Fig. 4: Charged particle trajectories after interaction with the outer hull of the spacecraft.

Our simulated pulsed power circuit delivers a sudden burst of current of 950 A for 5 ms within the power limitations of the ISS. The frequency and duty cycle of the pulses can be adjusted to avoid thermal overload of the coils. The pulses could produce a magnetic field strength up to 1.5 T in the centre of the coils. If sufficient warning had been provided of a solar particle event, ground personnel and crew would have time to react. The crew could enter the shielded module, whilst ground activated the shield, thus providing additional protection to crew for the duration of the increased solar activity. Table 2 summarises the parameters of the shield behaviour.

Table. 2: Calculated deflection of a 1 keV proton for a given current in a single coil of copper with resistance ~0.12 Ohm.

Current (A)	Voltage (V)	Magnetic Field (T)	Deflection angle (°)
50	6.0	0.072	8
640	76.8	1.000	59
6400	768	10.000	148

Future planned activities: 1U Sounding Rocket Payload

A payload has been designed and constructed (figure 2b) for launch on a MASER sounding rocket, reaching 250km altitude. The Swedish Space Corporation launch site in Kiruna is close to the Earth's magnetic north pole, providing a unique opportunity to expose experiments to very high doses of radiation. Our device will measure radiation with the shield off, and then switched on for a comparative data set.

The data collected during the shield-off period will also provide a great opportunity to study the Earth's near-space environment using many existing space observation platforms nearby; e.g., ionospheric radars, magnetometers and riometers (which remotely detect space radiation levels). The radiation data collected during this flight will be used to perform a new calibration of the Kiruna riometer, and compare with a calibration done using distant radio sources. Even if the launch occurs during quiet space weather conditions, the *in situ* radiation observations on our flight will provide a great opportunity to uncover new knowledge about the Earth's window into space - the ionosphere.

Conclusion

In spacecraft in Low Earth Orbit, astronauts are protected from radiation by the Earth's magnetic field, however, in lunar orbit and on the lunar surface, the radiation environment is significantly different and brings associated health risks. New technologies providing radiation shielding will be required for future long-term human spaceflight missions. Currently, all radiation protection is provided by passive shielding; the dimensions of which would need to be increased to provide the same protection in the higher radiation environment, however, increases in material thickness result in increased mass which becomes too heavy and bulky to be launched.

We have built two prototype active shields, from electromagnets, for the deflection of charged particles away from spacecraft. A key aspect of this design is the placement of electromagnetic coils in the walls of the spacecraft, creating a fully-protected void in the centre of any habitable modules. This ensures that crew are not exposed to high strength magnetic fields.

The prototype shields generate magnetic field strengths of < 10 mT when constructed from copper wire and supplied with currents typically found on board spacecraft. Simulations of these shield designs have shown that such a shield could deflect 1 keV protons by $\sim 8^\circ$ if able to produce a peak field strength of 72 mT. It may be possible to generate such a field by using a pulsed current to provide a short, rapid, high current. This would be of particular use for protecting crew against a known solar particle event.

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